

Chapter 14: Elements, Compounds and Mixtures

Textbook Exercise Solutions & Answers



1. Who are my companions?

Column 'A' Column B' 1. Stainless steel a. Non-metal 2. Silver b. Compound 3. Bhajani mixture for milling c. Mixture 4. Salt d. Element 5. Coal e. Alloy 6. Hydrogen f. Metal

Column 'A'	Column B'
1. Stainless steel	e. Alloy
2. Silver	f. Metal
3. Bhajani mixture for milling	c. Mixture
4. Salt	b. Compound
5. Coal	a. Non-metal
6. Hydrogen	d. Element

2. Write the names of elements from the following symbols:

Zn, Cd, Xe, Br, Ti, Cu, Fe, Si, Ir, Pt.

Write the names of elements from the following symbols: Zn, Cd, Xe, Br, Ti, Cu, Fe, Si, Ir, Pt.

Symbol	Element
Zn	Zinc
Cd	Cadmium
Xe	Xenon
Br	Bromine
Ti	Titanium
Cu	Copper
Fe	Iron
Si	Silicon
Ir	Iridium
Pt	Platinum

3. What are the molecular formulae of the following compounds?

(Hydrochloric acid, Sulphuric acid, Sodium chloride, Glucose, Methane)

What are the molecular formulae of the following compounds? (Hydrochloric acid, Sulphuric acid, Sodium chloride, Glucose, Methane)

- Hydrochloric acid $\rightarrow$ HCl
- Sulphuric acid $\rightarrow$ H₂SO₄
- Sodium Chloride $\rightarrow$ NaCl
- Glucose $\rightarrow$ C₆H₁₂O₆
- Methane $\rightarrow$ CH₄
- Water $\rightarrow$ H₂O
- Carbon dioxide $\rightarrow$ CO₂
- Sucrose (sugar) $\rightarrow$ C₁₂H₂₂O₁₁

4. Give scientific reasons:

(a) Buttermilk is churned to get butter.

1. Churning is the process of shaking up buttermilk to make butter.
2. Butter is essentially the fat of milk.
3. Churning physically agitates the cream until it ruptures the membrane surrounding milk fat
4. Fat droplets can join with each other to form clumps of fat.

(b) In Chromatography the ingredients of a mixture rise up to a limited height when water rises up to the upper end of the paper.

In Chromatography two properties of substances are used, (a) They are the stability of the substances in the solvent that moves up. The ability of the substance to stick to the stationary filter paper, (b) So all the components of the mixture do not rise all the way to the upper end of the filter paper but remain behind at limited heights.

(c) A wet cloth is wrapped around a water storage container in summer.

A wet cloth will absorb the heat from the surroundings and it will keep the water in the container, cool for a longer time.

5. Explain the difference.

(a) Metals and Non-metals

Metals	Non-metals
1. Metals are good conductors of heat and electricity.	1. Non-metals are poor conductors of heat and electricity.
2. Metals are solids at room temperature except for mercury.	2. Non-metals exists in all three states.
3. Metals are lustrous, (shiny)	3. Non-metals are not lustrous, (dull appearance) except graphite.
4. Metals are malleable (can be hammered into sheets)	4. Non-metals are brittle, not malleable.
5. Metals are ductile, can be drawn into wire.	5. Non-metals are not ductile.
6. Metals are very hard and strong.	6. Non-metals are brittle, will break down into pieces except diamond.

(b) Mixture and Compound

Mixture	Compound
1. The different substances are not chemically joined together to form a mixture but mixed physically.	1. A compound is formed when two or more elements chemically combine together e.g. $\text{H}_2\text{O} \rightarrow$ chemical formula of water.
2. Each substance in the mixture retains its own properties	2. Compound has fixed properties.
3. Mixtures are impure substances.	3. Compounds are pure substances.
4. Substances from the mixture can easily be separated by physical methods.	4. The constituents of a compound can be separated only by chemical methods.
5. The constituents of a mixture are present in varying proportions.	5. The constituents of a compound are present in fixed proportion.

(c) Atoms and Molecules

Atoms	Molecules
1. An atom is the fundamental part of matter.	1. A molecule is a group of chemically bonded atoms.
2. Atoms are basic building blocks of matter.	2. Molecule is the smallest unit of a chemical compound
3. It is foundation of molecules.	3. It is foundation of chemical compound.

(d) Separation by distillation and Separation by separating funnel

Separation by distillation	Separation by separating funnel
1. Distillation is used for purification of impure liquids, for separating liquid and solids (Separation of mixture containing two miscible liquids) e.g. to separate salt and water from saltwater.	1. This procedure is used for separating two liquids e.g. separating oil and water, (two immiscible liquids)

6. Write answers to the following questions in your own words.

(a) How are the components of mixture separated by simple methods?

1. Component of mixtures are separated by straining, filtering, sifting, picking, sorting, winnowing, combing with a magnet and sublimation.

2. In a mixture, the constituent substances do not lose their identity, they can be separated easily by physical methods.

a. Sedimentation: (i) It is a process of separating an insoluble solid from a liquid in which it is suspended, by allowing it to settle to the bottom of the container, e.g. Muddy water contains heavier particles like sand and soil, (ii) Leave this muddy water undisturbed for some time, (iii) The heavier soil and sand particles settle down and the clear upper water is poured out by decantation.

b. Filtration: It is used for separating insoluble solids from a liquid, e.g. mixture of chalk and water is poured through a filter paper in a funnel while the water gets

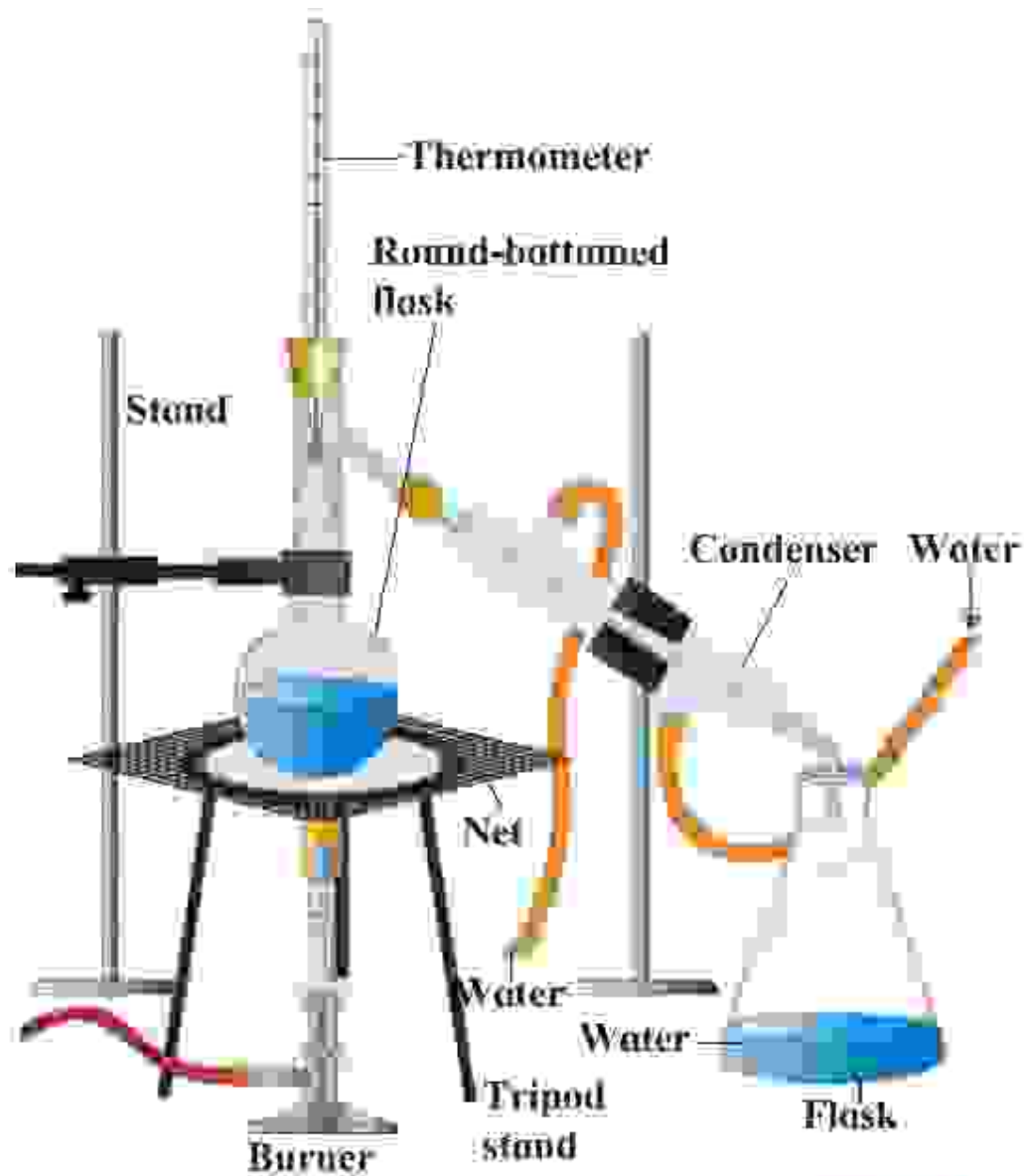


Filtration method

collected in the beaker below, chalk is retained in the filter paper.

c. Evaporation: It is used for recovering dissolved solid substances from solutions by evaporation e.g. sugar can be recovered from sugar-water.

d. Distillation: It is a process of heating a solution containing soluble solids to form vapours of the liquid and then cooling the vapours to get the liquid back. e.g. A mixture of common salt and water is taken in distillation flask and heated. Steam rises up and comes out into condenser.



Distillation method

e. Sublimation: It is a process in which some solids on heating are transformed directly to vapour without passing through the liquid state, (i) It is used to separate a mixture of solids. The vapours are cooled separately, (ii) Used to separate ammonium chloride, iodine, camphor and sulphur from any mixture.

f. Magnetic separation: A mixture with iron fillings as one of the components can be separated using magnet to attract the iron particles away from the mixture.

(b) Which elements (metals, non-metals) compounds and mixtures do we use in our day to day life?

Elements – non-metal

- Oxygen, nitrogen → present in air
- Hydrogen → present in water
- Silicon → in memory card
- Lithium → to make batteries
- Neon → in neon signs/lights
- Sulphur → used in water treatment, agricultural pesticides

Elements – metals

- Calcium → in milk
- Silver, gold → used in jewellery
- Aluminium, copper, iron → Kitchen vessels
- Mercury → in thermometer
- Copper → electric wires

Compounds:

- Sodium chloride → table salt
- Sodium carbonate → washing soda
- Sodium bicarbonate → baking soda
- Sodium hypochloride → bleaching powder
- Sodium hydroxide, Potassium hydroxide → in making soaps
- Calcium oxide, Calcium hydroxide → in white washing the buildings.
- Hydrochloric acid → in cleaning toilets
- Sucrose → sugar used in cooking and baking.

Mixtures:

- Bhel → mixture of puffed rice, sev, groundnuts, lemon juice
 - Concrete → mixture of cement, sand and rocks
 - Salad → mixture of onion, cucumber, tomato, lettuce, etc.
- Sherbet and saltwater are also mixtures.

(c) In everyday life, where and for what purpose do we use centrifugation?

Centrifugation: It is a process which involves application of centrifugal force.

1. It is used in industrial and laboratory settings. It is used for separation of fluids, gas or liquid based on density. In centrifugation mixture is separated through spinning and the solid settles to the bottom and the solution is clear.

2. Best example of centrifugal force is working of washing machine when it dries the clothes. The wet clothes are moving in circular path and a force acts on water particles in clothes and this force pulls water all outer side. Force involved is centrifugal force which removes the water from the clothes and clothes are dried in this way.

3. Cream separation: When the machine moves, the rod set inside milk moves and milk is pressurised and solid part from this milk goes outside due to centrifugal force and this solid part is cream. In this way the cream is separated from the milk.

(d) Where are methods of separation by distillation and by separating funnel used?

1. Distillation: It is the process of separating the components or substances from a compound. It is a process of heating a solution containing soluble solids to form vapours of the liquid and then cooling the vapours to get the liquid back.

Distillation method is used in the following:

- to separate components of air into oxygen, nitrogen, argon, etc.
- to separate salt from saltwater.
- to purify impure liquids.
- to prepare distilled water.

2. Separating funnel: When a mixture of two immiscible liquids is left undisturbed, two layers are clearly seen to have formed, (a) The heavier of the liquids remain below and the lighter liquid floats on it. (b) So two liquids in a mixture can be separated by making use of this property, e.g.

- to separate kerosene and water.
- to separate oil and water.

(e) Which precaution will you take while using the methods of distillation and separation by separating funnel?

In using distillation:

1. Allow sufficient space to work in, working area should be well lit and well ventilated to prevent the accumulation of alcoholic vapours.
2. Keep a fire extinguisher handy.
3. There should be no obstruction in the piping that could cause pressure build-up in the stills and cause it to burst.

In using separating funnel:

1. Stopper should be tightly fitted so that the solution does not leak out when the separating funnel is inverted.
2. Never throw any layer away until you are absolutely sure that you isolated the final product.